

KIRAN S BALIGA

+91 9544737577 · Kerala (Open to relocate) · kiransbaliga@gmail.com · [linkedin.com/in/kiransbaliga](https://www.linkedin.com/in/kiransbaliga) · baliga.dev

SUMMARY

Backend & AI Engineer with 3 years of experience specializing in production LLM reporting pipelines, Model Context Protocol (MCP) tooling, and distributed architectures using **Python**, **Node.js**, and **Ruby on Rails**. Experienced architecting asynchronous clinical AI pipelines, engineering fine-grained multi-tenant RBAC systems, and building behavioral upsell engines for **5M+** platform users. Available immediately.

SKILLS

Languages: TypeScript / JavaScript, Python, Ruby, SQL

Technologies: FastAPI, Django, Node.js, Ruby on Rails, MCP, LangChain, CrewAI, Kotlin Multiplatform, PostgreSQL, Docker, AWS, GraphQL

EXPERIENCE

Ultrahuman

Mar 2026 – June 2026

Backend Engineer — Remote · *Case Study:* baliga.dev/ultrahuman

- Architected and owned the end-to-end **LLM health-reporting pipeline** for Ultrahuman's performance lab, ingesting multi-modal diagnostics across **10+** stations (DEXA, VO2 max, APIs, PDFs) to extract biomarkers and synthesize **15-page** clinical plans for doctor review.
- Ruby backend batch processing through job queues caused delayed vital calculations. Ported the **SpO2 algorithm** to **Kotlin Multiplatform** for **on-device execution**, achieving **99%+** numerical parity on ring wearer data; pre-release benchmarks by engineering leads projected an **~80%** latency reduction.
- Enhanced the modular LLM translation pipeline for CJK languages via **trademark glossaries** and **context injection**, boosting quality from **86% to 91%** and cutting token costs by **~67%** through **prompt caching** and **request batching**.
- Revamped dynamic upsell system for subscription extensions (**Powerplugs**); evaluated biometric cohorts (heart rate, sleep scores) nightly across distributed daily batch queues (**5M+** users), driving **8k+** conversions in 2 months.

Keyvalue Software Systems

Sep 2023 – Feb 2026

Associate Software Engineer (Node.js, Python, React) — Kerala

Pencil — B2B AI Ads Platform

Case Study: baliga.dev/pencil

- Contributed to migrating large-scale workspaces from PostgreSQL to **Google Cloud SQL**, implementing schema changes and a **custom ORM** that improved API response times by **30%**.
- Engineered fine-grained **multi-tenant RBAC and tenant isolation**, implementing group/role hierarchy and composable boolean (AND/OR) permission checks across API endpoints.
- Migrated document exports from client React rendering to server-side **headless Puppeteer**. Resolved dynamic brand font injection and cross-device margin drift, making output consistent and **65%** faster.

Siren — Communication Rails for AI Agents

Case Study: baliga.dev/siren

- Designed and built core communication rails and message protocols powering **multi-agent orchestration** between autonomous AI agents.
- Built the **Siren Agent Toolkit** adhering to OpenAI tool specs for LangChain/CrewAI, and engineered a standalone **Model Context Protocol (MCP)** server that reduced developer setup effort by **~75%** (MCP server setup vs. manual SDK integration).
- Published packages to npm (@trysiren/agent-toolkit, @trysiren/mcp) and PyPI (siren-agent-toolkit), enabling rapid external integration across agent workflows.

Other Client Projects

- Delivered **Salesforce** synchronization under a one-month deadline by designing ingestion APIs, webhooks, and **eventual consistency** workflows, improving API performance by **12%**.

Tgh Tech

May 2022 – Oct 2022

Full Stack Engineer (Python, Flutter)

- Built a full-stack mental health platform with **Django**, implementing content sharing, anonymous therapist messaging, and appointment scheduling workflows.
- Built and deployed scalable backend services on **AWS**, ensuring reliable performance and secure user experiences.

Airpmo

Jun 2021 – Dec 2021

Flutter Developer (Part-time)

- Built key **Flutter** features for a construction SaaS platform, streamlining equipment, materials, personnel, and subcontractor operations across active job sites.

PROJECTS

Multi-Task Federated Learning for Deep Neural Networks (Final-Year B.Tech Project)

- Built an AI framework for **federated and multi-task learning**, enabling personalized model training while preserving data privacy across distributed clients. [GitHub](#)

EDUCATION

APJ Abdul Kalam Technological University

2019 – 2023

B.Tech, Computer Science and Engineering (Minor in ECE) — Kerala, India

- GPA: **8.43 / 10**